

Cleaning Up Light Pollution



Much of the exterior lighting installed on homes and apartment buildings does little or nothing to increase visibility, security, and safety, but it often unnecessarily lights up the neighbors' properties and the night sky. Unfortunately, current lighting practices also gobble up an excess of electricity while delivering poor lighting with glare, light trespass, and "up" light—light that is directed upward, where it can do nothing to aid pedestrians finding their way home at night. The organization I founded, the Dark Sky Society of Long Island, New York, provides information about how we can decrease the amount of light lost up into the night sky and how we can easily light the ground more effectively and efficiently.

At a recent monthly meeting of energy efficiency experts held at the Long Island Power Authority, I spoke about the environmental and financial costs of light pollution. At the end of my talk, one of the experts working with builders discussed a particular frustration she encounters on job sites. Because most energy efficiency programs do not include guidelines for outdoor lighting, she invariably finds that the builders implement all of the utility's recommendations inside the home, but continue to use energy-inefficient and light-polluting unshielded floodlights on the outside of the home.



(left) The Glarebuster fixture is easy for a homeowner or an electrician to install.
(right) Here is an example of a shield for existing floodlights.



The estimated yearly cost of light pollution in the United States exceeds \$4.5 billion for electricity, which is generated primarily by burning polluting fossil fuels. In addition to wasting energy, light at night negatively affects flora, fauna, and the health of humans and our environment. Numerous research studies have been conducted on the negative effects of light at night, proving that we need to change our outdoor lighting practices.

The negative effects on human health include sleep disturbance and suppression of the hormone melatonin from light in our bedrooms and coming through our windows. (Melatonin is not secreted in the presence of light.) Melatonin performs many important functions in the body, one of which is to inhibit tumor development. The negative effects on animals include the disruption of foraging, movement, pre-

reparation relationships, and circadian rhythms, which have evolved over millions of years in a day-night cycle. Untold numbers of migrating birds are thrown off course by bright lights. They can die from exhaustion or when they crash into lit buildings and towers. Research suggests that trees also suffer when they are lighted all night and do not respond to winter dormancy signals.

Fortunately, lighting manufacturers are developing dark-sky-friendly fixtures that reduce both light pollution and energy consumption. Organizations, including the Dark Sky Society, the Audubon Society, the Nature Conservancy, the Natural Resources Defense Council, and the Illuminating Engineering Society, among others, are providing guidelines and advice for more effective outdoor lighting that protects our health and environment.

Dark-sky fixtures, which include CFLs and shielding components, cost no more than traditional outdoor lighting fixtures. Most are just as easy to install and save homeowners money, often enough to pay for retrofits. For many applications, the improved fixtures can save up to 30% of the energy that unshielded fixtures use to light the same area, and do so

Fixture Choices for East Hampton, New York

The program to control light pollution in East Hampton, New York (see Figure) has been a success. Tara Burke, the senior planner for the East Hampton, says “Environmental affects and neighborhood compatibility are necessary elements of Site Plan review, which really can’t be analyzed completely without looking at outdoor lighting too. When ensuring that the outdoor lighting does not produce glare and ultimately is dark-sky friendly, the Planning Board can truly say that they have reviewed a site in its entirety and really feel good about the finished product.”

Figure. East Hampton businesses were given a choice of outdoor light fixtures, a few of which are illustrated on the right.



without glare. For example, if a homeowner replaces traditional security floodlights (two dusk-to-dawn 150W bulbs) with the equivalent light output in CFL bulbs and shielding, the savings will be at least \$100 per year, based on current utility prices on Long Island.

Simply changing the light bulb will help reduce light pollution, too. We would not have an unshaded bare bulb indoors because it creates glare. Glare is ugly and it reduces visibility. We need to shade outdoor bulbs, too. A 1,800 lumen (100W incandescent or 23W compact fluorescent) bare light bulb is too bright outdoors. Try a 500 lumen (40W incandescent or 9W compact fluorescent) bulb. An R-20 bulb (which has opaque sides) in a porch light will eliminate glare, and will direct all of the light downward.

Municipalities are taking the issue of outdoor lighting more seriously, not only to address residential complaints about light trespass, but also to improve the appearance and safety of the community at night. Tax dollars are saved when municipalities and schools use night lighting sensibly by turning off lights when they are not needed. Municipal planning departments now

require a lighting plan for new site plan submissions so they can better assess proposed exterior lighting for businesses. Fixtures are approved when they meet the municipal requirements for shielding, light levels, and mounting heights (see “Fixtures Choices for East Hampton, New York”).

Unfortunately, information for homeowners about good night lighting is still hard to come by, and dark-sky-friendly fixtures are not as widely carried in the big box stores and local hardware chains as fixtures that cause light pollution. Therefore, most lighting on the vast majority of homes—including million dollar mansions to apartment houses—consumes too much energy and performs poorly. This will change as more people become aware of the importance of controlling light pollution. It just makes good sense (and good neighbors).



—Susan Harder

Susan Harder lives in East Hampton, New York, and is the founder of the Dark Sky Society of Long Island, New York, with memberships in the International Dark Sky Association and the Illuminating Engineering Society.

FOR MORE INFORMATION:

See the findings in the pending outdoor lighting legislation for New York State at <http://assembly.state.ny.us/leg/?bn=S04474&sh=t>.

The Energy Federation, Incorporated, offers many choices for effective and dark-sky-friendly fixtures. Find information at www.energyfederation.org/consumer/default.php/cPath/25_1245.

One of these fixtures, the GlareBuster, is easy for the homeowner or an electrician to install. This fixture is available from ACE Hardware stores and through the Web at www.glarebuster.com.

For information on shielding existing floodlights, go to www.energyfederation.org/consumer/default.php/cPath/903.

To learn more about the dark-sky movement, go to www.darksky.org and www.darkskysociety.org.